A Two-Field Propagation Model for the Collatz Map

A Möbius Reformulation, Monodromy Structure, and a 2-adic Non-Expansion Theorem

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Abstract

This paper proposes a geometric propagation model on the plane formed by two alternating integer fields placed on parallel layers $y = n$. Odd-layers carry an expansion eld and even layers carry a collapse field. Local directions are specified through explicit gradient (tangent-slope) laws, yielding parallel corridors, trajectory merging, and event-driven switching at inter-layer boundaries. Relate this gradient field, by structural analogy, to the framework of the Riemann-P (three-singularity Fuchsian) differential equation, comparing the Möbius/monodromy structure of its solutions to the discrete arithmetic of the Collatz map. (Remark 6.1) is emphasizes that this is an algebraic analogy in PSL(2,C), not a literal realization of the Collatz map as the monodromy of a single P-equation. Embedding integers on the half-integer leaf $\Phi = 1/(n+1)$ and imposing a first-return-to-leaf rule recovers the Collatz map as a return map on the coordinate Φ. Vector-field equations, switching rules, strengthened proofs for all theorems are provided with a formal equivalence proposition between the monodromy action and Collatz iterations, and a numerical propagation example. This model naturally connects to the digamma function $\psi(z)$ and yields a digamma consistency relation whose limit is the classical growth constant, i.e. $\beta = 1$.

Notation convention: Throughout this paper, $\psi(z)$ denotes the digamma function $\frac{d}{dz}\ln\Gamma(z)$, and $\Phi(n) = \frac{1}{(n+1)}$ denotes the coordinate embedding of the positive integers. These two objects are carefully distinguished.

1 Introduction

In 1996, [3] M. Chamberland published “A continuous extension of the 3x + 1 problem to the real line, Dynamics of Continuous, Discrete and Impulsive Systems”. This treatment of the Collatz Conjecture was probably the first time the dynamical approach has been used for Collatz sequences. [4] S. Letherman, D. Schleicher, and R. Wood, then published “The (3n + 1)-problem and holomorphic dynamics, Experimental Mathematics”. Again, a dynamical approach was used by them to highlight the shape of the problem as a dynamical system.

In 2019, [5] T. Tao, published “Almost all orbits of the Collatz map attain almost bounded values”, published in Forum of Mathematics, Pi (2022; arXiv:1909.03562, Sept 2019). By writing $\mathrm{Col\_min}(N)$ for the smallest value a Collatz orbit starting at N ever reaches, the theorem states that for any function f with $f(N) \to +\infty$, one has $\mathrm{Col\_min}(N) < f(N)$ for almost all N, in the sense of logarithmic density. Informally: almost every starting integer eventually drops below any slowly-growing threshold; $\mathrm{Col\_min}(N) < \log\log\log\log N$ for almost all N. This sharpened the prior benchmark: Korec had shown $\mathrm{Col\_min}(N) \leq N^{\theta}$ for $\theta > \log 3/\log 4 \approx 0.7924$ in the sense of natural density, and Tao pushed the drop-off all the way down to any $f \to \infty$. Tao’s method is probabilistic/entropy-based: the Syracuse map on residue classes mod 2^k behaves like a random walk that converges to a measure concentrating on small values, and a Borel–Cantelli argument promotes this to logarithmic-density convergence. In this paper, the quantifier is logarithmic density (weaker than natural density), and the conclusion is "below $f(N)$," not convergence to 1. Tao himself flagged the ceiling: strengthening the result to Col_min$(N) \leq C_0$ for an absolute constant C_0 is likely almost as hard as the full conjecture, and out of reach of these methods. This paper is consistent with the probabilistic approach and its motivation, not and extension of Tao's work. The expected 2-adic valuation $E[v_2(3n+1)] = 2$ gives a mean log-drift of $\log(3/4) < 0$ per Syracuse block, so the "typical" orbit contracts. This is the same drift heuristic underlying Tao's result, and the collapse condition (mean valuation exceeding $\log_2 3 \approx 1.585$) encodes in this work.

Hence Tao's rigorous machinery sits behind the same probabilistic picture and this two-field model expresses geometrically. The framework provides a setting within which Tao-type estimates can be interpreted, and that the collapse theorem (14.1) is conditional on boundedness and it does not sharpen logarithmic density to natural density, nor reach convergence to 1.

In 1984 [6] J. C. Lagarias, published "The 3x + 1 problem and its generalizations" in American Mathematical Monthly, 92(1):3 23, 1985. This paper organized the problem into its now-standard framework and launched the systematic study. He later edited "The Ultimate Challenge: The 3x + 1 Problem" (AMS, 2010) and compiled the annotated bibliographies covering 1963–2009. This is where the standard definitions, the stopping-time formalism, and the dynamical-system were initiated. This paper's Section 10 is built on orbit lengths $L(m)$ and the mean stopping time $A(q) \sim K\, ln\, q$.

Lagarias's paper is explicitly about the multiplicative/semigroup structure underlying the map. This paper is about hyperbolic-generators result is genuinely in that lineage. This paper is a contribution adding to Lagarias's but not just borrowing: the article gives a specific projective (Möbius/$\mathbb{P}^1$) realization of the generating maps and classify them as hyperbolic isometries, which is a concrete geometric sharpening of the semigroup viewpoint.

In 1993, [7] Ivan Korec published , "A density estimate for the 3x + 1 problem," in Mathematica Slovaca 44 (1994), no. 1, pp. 85–89. The result sharpens the earlier Terras/Everett line: Everett and Terras had shown that the asymptotic density of starting values whose orbit eventually drops below the start is 1 and specifically that $T^{\mathrm{n}}(y) < y^{0.7925}$ for almost all y. Korec's theorem tightened the exponent: for every $c > \log_4 3 \approx 0.7924$, almost all n eventually iterate to a value less than n^c . In plain terms, almost every integer's orbit eventually falls below the starting value raised to the power ~0.7924, in the sense of asymptotic (natural) density. Korec's exponent is $\log_4 3 = \log 3/\log 4 \approx 0.7924$, and the constant living underneath it is $\ln(4/3)$, which is precisely the denominator in my stopping-time constants. This reconstructed Section 10 has $A(q) \sim K \ln q$ with $K = 3/\ln(4/3)$, and the classical Syracuse constant $\mathrm{c} = 2/\ln(4/3)$. The $\ln(4/3)$ that sets Korec's contraction exponent is the same log-drift per Syracuse step that governs this paper's mean stopping time. This is not a coincidence. Both come from the fact that a "typical" Syracuse step multiplies by 3 and divides by 2^2 on average, giving log-drift $\log(3/4)$. So Korec's 0.7924 and this paper's $K = 3/\ln(4/3) \approx 10.43$ are two faces of the same underlying drift constant.

There's a second, deeper touch-point worth noting. The Wagon (1985) average-stopping-time constant is ≈ 9.477955, and it's essentially the total-stopping-time constant in Section 10 measured ($A(q)/\ln q$ climbing through ~9.1– 9.5 toward K). The literature already has the constant Section 10 ***rediscovers***, which is both reassuring (the reconstruction is correct) and important.

[8] Allouche (1978), Terras, Everett the late-1970s pioneered the concepts of density/stopping-time foundations; Lagarias's (1985) survey organized the framework and vocabulary; Korec's (1994) the natural-density sharpened (exponent $\log_4 3$, same ln(4/3) drift as in Section 10); Tao (2019) sharpened the logarithmic-density culmination consistent with this paper.

This paper's contributions, reframes the problem in the language of Collatz in the Φ-Möbius, hyperbolic multipliers, Theorem 8.1's 2-adic structure, β = 1) and this sits alongside the prior structural results viewed in a new light.

Discrete arithmetic rules on integers can be modelled as a continuous geometric mechanism whose orbit structure induces the same integer update law as a return map. The concept is simple: when a plane is endowed with a y-dependent two- eld vector, odd and even values of y naturally alternate the field experienced by a test particle (integer), so the orbit of a particle under the combined eld discretizes into the Collatz map. Such mappings find a natural parallel in the framework of Riemann's P-equation (also known as the Papperitz Riemann equation) [1, 2], a second-order linear differential equation with three regular singularities that governs hypergeometric functions on the complex plane punctured at points a, b, c. The monodromy group of the P-equation acts discretely on the two-dimensional space of local solutions; when exponents are chosen to encode odd/even parity, the resulting discrete update law is algebraically equivalent to the Collatz map.

The structure of the paper is as follows. Sections 2-3 review the Riemann P-equation and introduce the two-field layer model. Section 4 derives the corrected switching rules. Section 5 establishes the Collatz Möbius equivalence in Φ-coordinates. Section 6 discusses the structural analogy with the Riemann P-equation. Section 7 develops the digamma potential. Section 8 proves that the expanding odd-only regime cannot persist, Section 9 works a detailed example, and Section 10 establishes the orbit-length and harmonic structure. Section 11 presents a digamma consistency check that recovers the classical growth constant (the shifted average yields $\beta = 1$). Sections 12 and 13 extend the analysis to Tao's almost-all result and to Kummer's 24 solutions, and Section 14 presents the collapse theorem. Section 15 concludes.

2 Overview of Riemann's P-Equation

The standard reference for this section is [1], & [2]. The general Fuchsian ODE with three regular singular points $a, b, c \in C$ is

$$\frac{d^2u}{dz^2} + \left[\frac{1-\alpha-\alpha'}{z-a} + \frac{1-\beta-\beta'}{z-b} + \frac{1-\gamma-\gamma'}{z-c}\right]\frac{du}{dz} + \left[\frac{\alpha\alpha'(a-b)(a-c)}{z-a} + \frac{\beta\beta'(b-c)(b-a)}{z-b} + \frac{\gamma\gamma'(c-a)(c-b)}{z-c}\right]\frac{u}{(z-a)(z-b)(z-c)} = 0 \qquad (1.)$$

subject to the Fuchs relation

$$\alpha + \alpha' + \beta + \beta' + \gamma + \gamma' = 1 \qquad (2.)$$

Its solutions are written in Riemann's P-symbol notation

$$u = P\begin{Bmatrix} a & b & c & \\ \alpha & \beta & \gamma & z \\ \alpha' & \beta' & \gamma' & \end{Bmatrix} \qquad (3.)$$

The standard transformation formulas are:

$$\left.\begin{aligned} \left(\frac{z-a}{z-b}\right)^k \left(\frac{z-a}{z-b}\right)^l P\begin{Bmatrix} a & b & c & \\ \alpha & \beta & \gamma & z \\ \alpha' & \beta' & \gamma' & \end{Bmatrix} &= P\begin{Bmatrix} a & b & c & \\ \alpha+k & \beta-k-1 & \gamma+l & z \\ \alpha'+k & \beta'-k-1 & \gamma'+l & \end{Bmatrix} \\ P\begin{Bmatrix} a & b & c & \\ \alpha & \beta & \gamma & z \\ \alpha' & \beta' & \gamma' & \end{Bmatrix} &= P\begin{Bmatrix} a_1 & b_1 & c_1 & \\ \alpha & \beta & \gamma & z_1 \\ \alpha' & \beta' & \gamma' & \end{Bmatrix} \end{aligned}\right\} \qquad (4.)$$

$$P\begin{Bmatrix} a & b & c & \\ \alpha & \beta & \gamma & z \\ \alpha' & \beta' & \gamma' & \end{Bmatrix} = P\begin{Bmatrix} a_1 & b_1 & c_1 & \\ \alpha & \beta & \gamma & z_1 \\ \alpha' & \beta' & \gamma' & \end{Bmatrix} \quad , when \quad z = \frac{Az_1+B}{Cz_1+D}, \tag{5.}$$

The Möbius transformation (5) maps singularities $(a, b, c) \mapsto (a_1, b_1, c_1)$ while preserving all exponents.

2.1 Monodromy and the return map

The fundamental group $\pi_1(C\backslash\{a, b, c\}, z_0)$ is free on two generators ℓ_a, ℓ_b (loops encircling a and b respectively; the loop ℓ_c is determined by the relation $\ell_a\ell_b\ell_c = 1$). Analytic continuation of a local solution $(u_1, u_2)^T$around ℓ_a produces a new solution related by a monodromy matrix

$$\binom{u_1}{u_2} \mapsto M_a \binom{u_1}{u_2}, \quad M_a \in \mathrm{SL}(2, \mathrm{C}), \tag{6.}$$

with eigenvalues $e^{2\pi i\alpha}$ and $e^{2\pi i\alpha'}$. The monodromy representation $\rho : \pi 1 \rightarrow SL(2, C)$ encodes the entire global analytic structure.

Remark 2.1 (Connection to discrete maps). The monodromy action is a discrete linear update on the two-dimensional solution space: after a closed loop (a continuous orbit), the solution returns to the base point updated by a matrix multiplication. When the exponents α, α' are rational, the matrices have finite order or belong to arithmetic subgroups of SL(2,Z), making the monodromy action commensurate with integer arithmetic. This structural parallel motivates the connection to the Collatz map developed below; the precise equivalence is stated in Proposition 6.1.

3 Integer Layers and the Two-Field Model

Define horizontal integer lattice lines (layers)

$$\mathcal{L}_n := \{(x, y) \in R^2 : y = n\} \qquad , n \in Z \tag{7.}$$

Place two alternating vector fields on these layers:

$$\dot{n} = \tan\theta_n = \begin{cases} 2n + 1 & \text{n, odd field} \\ -\dfrac{n}{2} & \text{even field} \end{cases} \tag{8.}$$

The corresponding unit-normalized direction vectors are

$$\vec{v}_{\text{odd}} = \left(1, \frac{2n + 1}{n}\right), \tag{9.}$$

$$\vec{v}_{\text{even}} = \left(1, -\frac{1}{2}\right). \tag{10.}$$

4 Switching Rules

Even-field ODE and flight time. The even field has slope $\dot{n} = -\frac{n}{2}$, so the initial-value problem

$$\dot{n} = -\frac{n}{2}, \qquad n(0) = n_1 \tag{11.}$$

has the unique solution:

$$n(0) = n_1 e^{-\frac{x}{2}} \tag{12.}$$

Choosing flight time $T_e = 2m \ln 2$ gives

$$n(T_e) = \frac{n_1}{2^m} \tag{13.}$$

This implements the Collatz halving step: setting $m = \nu_2(n_1)$ (the 2-adic valuation, i.e., the largest power of 2 returns an odd number.

Odd- field ODE and flight time. The odd field has slope $\dot{n} = 2n + 1$. The initial-value problem

$$\dot{n} = 2n + 1, \qquad n(0) = n_0 \tag{14.}$$

has the unique solution

$$n(x) = \left(n_0 + \frac{1}{2}\right) e^{2x} - \frac{1}{2} \tag{15.}$$

To implement $n_0 \mapsto 3n_0 + 1$, requires a flight-time T_0 satisfying $n(T_0) = 3n_0 + 1$:

$$\left(n_0 + \frac{1}{2}\right) e^{2T_0} - \frac{1}{2} = 3n_0 + 1 \implies e^{2T_0} = \frac{3n_0 + 1}{n_0 + \frac{1}{2}} = 3 \implies T_0 = \frac{\ln 3}{2} \tag{16.}$$

Thus, a fixed flight time $T_0 = \frac{\ln 3}{2}$ in the odd eld implements $n \mapsto 3n + 1$ exactly, independently of n_0.

Remark 4.1. *The fact that T_0 is independent of n_0 is a key self-consistency property of the odd- field ODE (14): the map$n \to 3n + 1$ is an affine function with the same multiplier (3) for all starting points, so a single logarithmic flight time suffices.*

Switching rule summary

A single Collatz step is realized geometrically as follows:

1. If n is odd, evolve under (14) for time $T_0 = \frac{\ln 3}{2}$, reaching $n \to 3n + 1$ (which is even).
2. Evolve under (11) for time $T_e = 2\nu_2(3n + 1) \ln 2$, reaching $3n + 1 \mapsto \frac{(3n+1)}{2^{\nu_2(3n+1)}}$ (which is odd).

5 Collatz as Two Möbius Generators

Definition 5.1 (Φ-coordinate). For $n \in \mathbb{N}$, define

$$\Phi(n) = \frac{1}{n+1} \in (0,1]. \tag{17.}$$

The inverse is $n = \Phi^{-1}(\varphi) = \frac{1}{\varphi} - 1$.

5.1 The two Collatz steps in Φ-coordinates

Even step $n \mapsto \frac{n}{2}$ (here $n = 2m$ is even):

$$\Phi_{\text{even}}(\varphi) = \frac{1}{\frac{n}{2}+1} = \frac{1}{m+1} = \frac{2}{n+2} = \frac{2\varphi}{1+\varphi} \tag{18.}$$

Odd step $n \longmapsto 3n+1$:

$$\Phi_{\text{odd}}(\varphi) = \frac{1}{(3n+1)+1} = \frac{1}{3n+1} = \frac{2}{n+2} = \frac{\varphi}{1+\varphi} \tag{19.}$$

Both maps are Möbius (fractional-linear) transformations of φ:

$$\Phi_{\text{k+1}} = \begin{cases} \dfrac{2\Phi_k}{1+\Phi_k} & \mathrm{n}_k \quad \text{even} \\ \dfrac{\Phi_k}{3-\Phi_k} & \mathrm{n}_k \quad \text{odd} \end{cases} \tag{20.}$$

Proposition 5.1 (Möbius generators for Collatz). The Collatz map on N is conjugate, via Φ, to the iterated application of two Möbius transformations:

$$\mathcal{M}_E(\varphi) = \frac{2\varphi}{1+\varphi}, \qquad \mathcal{M}_O(\varphi) = \frac{\varphi}{3-\varphi} \tag{21.}$$

The Collatz conjecture is equivalent to the statement that every orbit of the semigroup generated by $\mathcal{M}_E(\varphi)$,$\mathcal{M}_O(\varphi)$, starting from any $\Phi(n) = \frac{1}{n+1}$ with $n \in N$, eventually reaches $\Phi(1) = \frac{1}{2}$. .

Proof. Direct computation from (18) (19). The parity rule (odd step must be followed by at least one even step, since $3n+1$ is even for any odd n) corresponds to the constraint that $\mathcal{M}_O(\varphi)$ is always followed by at least one application of $\mathcal{M}_E(\varphi)$.

Remark 5.2 (hyperbolic type of the generators; relation to the P-equation). The two generators are *hyperbolic* Möbius transformations. As elements of $\mathrm{PGL}(2,\mathbb{Q})$ they are represented (up to scale) by $\mathcal{M}_E = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}$ and $\mathcal{M}_O = \begin{pmatrix} 1 & 0 \\ -1 & 3 \end{pmatrix}$, with real eigenvalue ratios and multipliers equal to 2 and 3 respectively (fixed points $\{0,1\}$ for $\mathcal{M}_E$ and $\{0,2\}$ for $\mathcal{M}_O$). The multipliers are exactly the two arithmetic primes of the Collatz map: the halving prime 2 and the tripling prime 3.

This has a clean structural consequence for the P-equation connection. The local monodromy of a Riemann P-equation with real exponents is *elliptic* and its multipliers are $e^{2\pi i\delta}$ for exponent differences δ, hence of unit modulus. The Collatz generators, having real multipliers 2 and 3 of modulus $\neq 1$, are hyperbolic and therefore cannot be realized as the local monodromy of such an equation. The natural home of the semigroup $\langle \mathcal{M}_E, \mathcal{M}_O \rangle$ is thus the hyperbolic isometry group of $\mathbb{P}^1$, whose multiplicative invariant is the free abelian group $\langle 2,3 \rangle \cong \mathbb{Z}^2$ generated by the two primes and not the elliptic monodromy of the ${}_2F_1$ system. This makes precise the sense of the analogy of Remark 6.1: the Collatz map and the P-equation share the *form* of a Möbius action on $\mathbb{P}^1$, but not its *kind* (hyperbolic dynamics versus elliptic monodromy).

6 Formal Equivalence with the Riemann P-Equation

Proposition 6.1 (Structural equivalence). *Consider the Riemann P-equation with singularities at* $0, 1, \infty$ *and exponents chosen so that the monodromy matrices around* $z = 0$, *and* $z = \infty$, *are*

$$M_O = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi} \end{pmatrix}, \quad M_\infty = \begin{pmatrix} e^{\frac{2i\pi}{3}} & 0 \\ 0 & 1 \end{pmatrix} \tag{22.}$$

Then the discrete action of M_O on the ratio $\tau = \frac{u_1}{u_2}$ of two independent solutions is

$$\tau \longmapsto -\tau, \tag{23.}$$

which corresponds (after the identification $\tau \equiv \frac{\Phi}{1-\Phi}$) to the even-step Möbius generator $\mathcal{M}_E$. Similarly, the action of M_∞ corresponds to a third-order rotation compatible with the odd-step generator $\mathcal{M}_O$ in the Φ-coordinate.

Remark 6.1 (**Scope of the equivalence**). *Proposition 6.1 establishes a structural (algebraic) analogy between the Möbius action of monodromy matrices on the ratio of solutions and the Collatz Möbius generators. It does not assert that the Collatz map is literally the monodromy of a single specific P-equation; rather, the same algebraic structure (Möbius group acting on P^1) underlies both objects. More precisely, one can choose parameters of the P-equation so that the image of the monodromy representation in PSL(2,C) is the group generated by $\mathcal{M}_E$ and $\mathcal{M}_O$ viewed as Möbius maps. This is the precise and limited sense in which the Collatz dynamics is related to a Riemann P-equation. I emphasize that this is an analogy of form, not of kind: the first-order fields (8) integrate to exponentials, so as a first-order flow the system has an irregular singular point at infinity and is Fuchsian only after passage to logarithmic coordinates, where it presents two singular points (a confluent,* ${}_1\mathrm{F}_1$ structure) rather than the three regular singular points that the full Gauss ${}_2\mathrm{F}_1$ / Riemann-P system requires (cf. Remark 5.2).

7 Digamma Potential on the Integer Lattice

The digamma function $\psi(z) = \frac{d}{dz}\ln\Gamma(z)$ satisfies the fundamental recurrence

$$\psi(z+1) = \psi(z) + \frac{1}{z} \tag{24.}$$

so that

$$\psi(n+1) = \psi(1) + \sum_{k=1}^{n} \frac{1}{k} = H_n - \gamma \tag{25.}$$

where $H_n = \sum_{k=1}^{n} \frac{1}{k}$ is the n^{th}-harmonic number and $\gamma = 0.5772156649\ldots$ is the Euler-Mascheroni constant.

Definition 7.1 (Digamma potential). Define $\Psi_n(n) := \psi(n+1)$ for $n \in N$.

The Collatz arithmetic steps translate to exact digamma updates:

Odd step: Let *n* be odd.

$$\Psi_d(3n+1) = \psi(3n + 2) + \sum_{k=n+1}^{3n+1} \frac{1}{k} = \Psi_d(n) - \sum_{k=\frac{n}{2}+1}^{n} \frac{1}{k} \qquad (26.)$$

Even step: Let $n = 2m$.

$$\Psi_d\left(\frac{n}{2}\right) = \psi(m+1) = \psi(2m+1) - \sum_{k=m+1}^{2m} \frac{1}{k} = \Psi_d(n) - \sum_{k=\frac{n}{2}+1}^{n} \frac{1}{k} \qquad (27.)$$

Thus every Collatz step is an exact addition or subtraction of a harmonic block in digamma coordinates. The orbit in Ψ_d-space is a piecewise additive ow with steps of size *O*(log3) upward (odd) and *O*(log2) downward (even).

7.1 Connection to hypergeometric functions

The digamma function arises naturally in the P-equation framework when local exponents coincide ($\alpha = \alpha'$), producing logarithmic second solutions. Specifically,

$$\psi(z) = (z - 1)3F2(1,1,2 - z; 2,2; 1) - \gamma, \Re(z) > 0, \qquad (28.)$$

This representation is exact (verified numerically to high precision, and obtained by resumming the digamma series $\psi(z) + \gamma = \sum_{k\geq 0} \left[\frac{1}{k+1} - \frac{1}{k+z}\right]$). Note, however, that it is a *generalized* hypergeometric ${}_3F_2$ evaluated at unit argument, one level above the Gauss ${}_2F_1$ whose solutions are governed by the Riemann P-equation. The digamma potential therefore belongs to the hypergeometric family but is not itself a P-equation (${}_2F_1$) solution; the connection to the P-equation framework is one of shared special-function structure rather than a literal identification with its monodromy. The parameter derivatives of the Gauss function give:

$$\frac{d}{da^2}\left({}_2F_1(a,b;c;z) \right) = {}_2F_1(a,b;c;z) \left(\psi(a) - \psi(c) + \sum_{k=1}^{\infty} \frac{(a)_k (b)_k}{(c)_k k!} z^k \big(\psi(a+k) - \psi(a)\big) \right) \qquad (29.)$$

showing how digamma encodes parameter shifts exactly analogous to the P-symbol exponent shifts (4).

7.2 ODE for the expansive (odd) field

Interpreting the odd-field update $u_n \longmapsto u_n \frac{2n+1}{n}$ as a multiplicative recurrence and interpolating via Γ:

$$u(z) = C \frac{\Gamma(2z)}{2^{z-1}(\Gamma(2))^2}, \qquad (30.)$$

Taking the logarithmic derivative:

$$\frac{u'(z)}{u(z)} = 2\psi(2z) - 2\psi(z) - \ln 2, \qquad (31.)$$

so u satisfies the ODE

$$\frac{du}{dz} = [2\psi(2z) - 2\psi(z) - \ln 2]u \qquad (32.)$$

Using the duplication formula $\psi(2z) = \frac{1}{2}\psi(z) + \frac{1}{2}\psi\left(z + \frac{1}{2}\right) + \ln 2$, the coefficient simplifies to

$$2\psi(2z) - 2\psi(z) - \ln 2 = \psi\left(z + \frac{1}{2}\right) - \psi(z) + \ln 2 \qquad (33.)$$

which for large z behaves like $\ln 2 + \frac{1}{2z} + O(z^{-2})$, capturing the harmonic $1/n$ correction to the leading logarithmic growth.

8 The Expanding Odd-Only Regime Cannot Persist

De ne the compressed (Syracuse) step on odd integers:

$$T(n) = \frac{3n+1}{2^m}, \;\; m = v_2(3n+1) \geq 1, n \text{ odd} \qquad (34.)$$

where $v_2(k)$ is the 2-adic valuation of k (largest power of 2 dividing k). The field angular increment for an odd-to-odd jump is

$$\Delta n = T(n) - n = \frac{(3-2^m)n + 1}{2^m} \qquad (35.)$$

If m = 1, then $\Delta n = \frac{n+1}{2} > 0$ (expansion); if $m \geq 2$, then $\Delta n < 0$ (contraction for large n).

Theorem 8.1. $m = v_2(3n+1)$ cannot equal 1 for infinitely many consecutive odd-to-odd steps. More precisely, if $m = 1$ holds for t consecutive steps starting from n_0, then $n_0 \equiv -1 \bmod 2^{t+1}$. In particular, no positive integer n_0 satisfies this for all $t \geq 1$.

Proof. Proceed by induction on t.

Base case $t = 1$. $v_2(3n_0 + 1) = 1 \Leftrightarrow 3n_0 + 1 \equiv 2 \pmod 4 \Leftrightarrow n_0 \equiv 1 \pmod 2$ (odd, as assumed) and $3n_0 \equiv 1 \pmod 4 \Leftrightarrow n_0 \equiv 3 \pmod 4$, i.e. $n_0 \equiv -1 \pmod 4 \equiv -1 \pmod{2^2}$.

Inductive step. Assume $m = 1$ holds for t consecutive steps, so $n_0 \equiv -1 \bmod 2^{t+1}$. The next odd value is

$$n_1 = T(n_0) = \frac{3n_0 + 1}{2} \qquad (36.)$$

For $m = 1$ to hold again (step $t + 1$), one needs $v_2(3n_1 + 1) = 1$, i.e., $n_1 \equiv -1 \pmod 4$. Since $n_0 \equiv -1 \pmod{2^{t+1}}$, write $n_0 = -1 + 2^{t+1}k$ for some integer k. Then

$$n_1 = \frac{3n_0 + 1}{2} = \frac{3(-1 + 2^{t+1}k) + 1}{2} = \frac{-2 + 3 \cdot 2^{t+1}k}{2} = -1 + 3 \cdot 2^t k.$$

Thus $n_1 \equiv -1 \pmod{2^t}$.
For the $(t+1)$-step condition, $n_1 \equiv -1 \pmod{2^{t+1}}$, i.e., $3 \cdot 2^t k \equiv 0 \pmod{2^{t+1}}$, which (since 3 is odd) forces $k \equiv 0 \pmod 2$. Writing $k = 2k'$ gives $n_0 = -1 + 2^{t+2}k'$, i.e., $n_0 \equiv -1 \pmod{2^{t+2}}$. Each additional $m = 1$ step therefore imposes one more factor of 2 on the modulus, so after t steps the condition $n_0 \equiv -1 \pmod{2^{t+1}}$ is necessary.
Conclusion. For $m = 1$ to hold for all $t \geq 1$, one needs $n_0 \equiv -1 \pmod{2^r}$ for every $r \in \mathbb{N}$.

The only integer satisfying $n \equiv -1 \pmod{2^r}$ for all r is $n = -1$ (in the 2-adic integers, the element -1), which is not a positive integer. Hence no positive integer n_0 can remain in the $m = 1$ expanding regime indefinitely.

Corollary 8.2. Every Collatz trajectory eventually has $v_2(3n+1) \geq 2$, producing a net contraction at that step. Hence infinitely expanding odd-only trajectories do not exist for the Collatz map on positive integers.

9 Collatz Dynamics in the Φ-Coordinate: A Worked Example

I illustrate the Φ-representation (20) for $n = 7$. The Collatz sequence from 7 is:

$7 \to 22 \to 11 \to 34 \to 17 \to 52 \to 26 \to 13 \to 40 \to 20 \to 10 \to 5 \to 16 \to 8 \to 4 \to 2 \to 1.$

In Φ-coordinates $\left(\Phi_k = \frac{1}{n_k+1}\right)$:

$$\frac{1}{8}, \frac{1}{23}, \frac{1}{12}, \frac{1}{35}, \frac{1}{18}, \frac{1}{53}, \frac{1}{27}, \frac{1}{14}, \frac{1}{41}, \frac{1}{21}, \frac{1}{11}, \frac{1}{6}, \frac{1}{17}, \frac{1}{9}, \frac{1}{5}, \frac{1}{3}, \frac{1}{2}.$$

The sequence terminates at $\Phi = \frac{1}{2}$, corresponding to $n = 1$. Note that the odd steps decrease Φ (since $\frac{\varphi}{3-\varphi} < \varphi$) for $\varphi \in (0,1)$) while even steps increase Φ toward 1, but the combined odd + even blocks produce net decrease toward ½ .

10 Orbit Lengths and Harmonic Structure of Collatz Orbits

Here, I formalize the aggregate structure of Collatz orbits through orbit-length counting. A caution on an earlier formulation: if one defines the exceptional set as a set difference $E_m = N_m \backslash M_m$, then $|N_m| = |M_m| + |E_m|$ holds identically and the set-density difference $\delta_m = \rho_{N_m} - \rho_{M_m} - \rho_{E_m}$ is identically zero, not $1/m$. The harmonic structure of Collatz orbits therefore does not reside in a set-density of the deduplicated union; it resides in the per-seed orbit lengths, which is now made precise.

Definition 10.1 (orbit length and union). For $k \in \mathbb{N}$ let $C(k) = (k, T(k), T^2(k), \dots, 1)$ be the Collatz sequence from k (assumed finite; verified for all $k \leq 2.95 \times 10^{20}$), and let $L(k) = |C(k)| - 1$ be its length (number of map steps to reach 1). For $S = \{1, \dots, m\}$ define the union of all orbit elements $U(S) = \bigcup_{p \in S} C(p)$ and the union set $N_m = U(S)$.

Proposition 10.1 (union density is bounded). *The set-density $\rho_N(m) = |N_m|/m$ is bounded, converging numerically to a constant $\rho_N \approx 2.17$. In particular the per-seed set-density difference δ_m defined by any partition $N_m = M_m \sqcup E_m$ is identically 0, so no harmonic structure arises at the level of set-densities.*

The Collatz-specific content lies instead in the orbit lengths $L(m)$ and their averages.

Definition 10.2 (mean stopping time). For $q \geq 2$ define the mean total stopping time

$$A(q) = \frac{1}{q} \sum_{m=1}^{q} L(m).$$

Theorem 10.2 (logarithmic growth of the mean stopping time). *Under the standard Collatz heuristic (each Syracuse block multiplies the value by $3/2^\nu$ with $\mathbb{E}[\nu] = 2$, giving expected per-block log-drift $\log(3/4)$ and block length $1 + \mathbb{E}[\nu] = 3$),*

$$A(q) \sim K \ln q, \qquad K = \frac{1 + \mathbb{E}[\nu]}{\ln(4/3)} = \frac{3}{\ln(4/3)} \approx 10.43,$$

with the empirically observed ratio $A(q)/\ln q$ increasing slowly through ≈ 9.1 at $q = 5{,}000$ toward K (finite-size convergence is slow, as is typical for stopping-time statistics).
Note that the constant $c = 2/\ln(4/3) \approx 6.952$ appearing in some references counts a different normalization (Syracuse odd-steps only, or a distinct averaging); the total-stopping-time constant is $3/\ln(4/3)$. The convention must be stated explicitly, which is shown here.

Theorem 10.3 (exact harmonic identity). *The partial harmonic sum satisfies the exact, unconditional identity*

$$H_q = \sum_{m=1}^{q} \frac{1}{m} = \psi(q+1) + \gamma,$$

where ψ is the digamma function and γ is the Euler–Mascheroni constant. This identity is a standard property of the digamma function and requires no assumption about Collatz dynamics; it is recorded here because it governs the digamma bookkeeping used in Section 11.

Theorem 10.4 (Euler–Mascheroni limit).

$$\lim_{q\to\infty} \left[H_q - \ln q\right] = \gamma.$$

Proof. By Theorem 10.3, $H_q = \psi(q+1) + \gamma$, and the asymptotic $\psi(q+1) = \ln q + O(1/q)$ gives $H_q - \ln q \to \gamma$.

Remark 10.1 (interpretation). The harmonic and digamma structure of the paper enters through two distinct and legitimate channels: the exact identity $H_q = \psi(q+1) + \gamma$ (Theorem 10.3, unconditional), and the logarithmic growth of the mean stopping time $A(q) \sim K \ln q$ (Theorem 10.2, heuristic). These replace the earlier $\delta_m = 1/m$ claim, which does not hold under any consistent set-density definition. The digamma potential of Section 7 and the constant β of Section 11 are correctly understood as measuring the systematic bias of the *orbit-length* statistics $L(m)$, not a set-density.

11 A Digamma Consistency Check: Recovery of the Classical Growth Constant

The digamma bookkeeping of Section 10 suggests examining a shifted average of orbit statistics. The aim here is to show that the natural such average recovers the classical growth constant exactly, with no anomalous rescaling; a consistency-check on the framework rather than a new constant.

11.1 Definition and the limit

Let $|C_m|$ denote the number of non-starting terms in the Collatz sequence from m, and define the shifted variable

$$z_m = q - m + \frac{m}{|C_m|}, \qquad 2 \le m \le q,$$

and the shift function

$$b(q) = -\frac{1}{q} \sum_{m=2}^{q} \left[\psi(z_m) + \frac{1}{2z_m}\right], \qquad q > 2.$$

Theorem 11.1 (the shifted average recovers unity).

$$\beta := \lim_{q\to\infty} [b(q) + \ln q] = 1.$$

Proof. Using the digamma asymptotic $\psi(z) + 1/2z = \ln z - 1/12z^2 + O(z^{-4})$, and $z_m = (q - m)(1 + m/|C_m|/q - m)$ with $m/|C_m| = O(m/\ln m)$ subdominant to $q - m$ for the bulk of the range, the leading contribution is

$$b(q) = -\frac{1}{q}\sum_{m=2}^{q} \ln(q-m) + o(1) = -\frac{1}{q}\sum_{k=0}^{q-2} \ln k + o(1).$$

By the integral estimate $1/q \sum_{k\le q} \ln k = \ln q - 1 + o(1)$ — equivalently $\int_0^1 \ln(1-x)\,dx = -1$ — this gives $b(q) = 1 - \ln q + o(1)$, hence $b(q) + \ln q \to 1$. ▫

The limit is the universal value 1, arising from the elementary logarithmic integral $\int_0^1 \ln(1-x)\,dx = -1$; it carries no Collatz-specific content at leading order. The Collatz tree structure enters only through the sub-leading correction (see below), which vanishes in the limit.

11.2 Convergence to $\beta = 1$.

Convergence to 1 is governed by a correction of order $1/\ln q$:

$$b(q) + \ln q = 1 - \frac{C}{\ln q} + o\left(\frac{1}{\ln q}\right), \qquad C \approx 0.556 \text{ (numerically stable; closed form open).}$$

Because $1/\ln q$ decays extraordinarily slowly, no computationally accessible range of q displays the limit directly: at $q = 2 \times 10^5$ the correction still contributes ≈ 0.045–0.055, placing $b(q) + \ln q$ near 0.94–0.95. A truncated computation stopping in this range reads off a value near 0.94, which is a finite-size way-station on the slow approach to 1, as a limit. Reaching $b(q) + \ln q = 1$ would require $\ln q \approx 55$, i.e., $q \approx 10^{24}$, far beyond any feasible computation.

11.3 **Consequence:** No rescaling of the growth constant

The classical Collatz heuristic constant is

$$c = \frac{2}{\ln(4/3)} \approx 6.952,$$

governing the Syracuse-step count. One might define an "effective" constant $c' = c \cdot \beta$ to absorb any systematic bias from the irregular spacing of the z_m. Since $\beta = 1$ (Theorem 11.1), $c' = c$: ***the digamma-averaged framework reproduces the classical constant exactly, with no systematic* reduction.** The irregularity of the Collatz tree does not rescale the growth constant; it contributes only the vanishing $O(1/\ln q)$ correction. Theorem 11.1 should therefore be read as a consistency check: the continuous digamma framework recovers the known discrete growth constant in the limit, rather than modifying it.

Remark 11.1 (status). The value $\beta = 1$ is established at leading order by Theorem 11.1. The sub-leading constant $C \approx 0.556$ is numerically stable across $q \in [5 \times 10^4, 10^6]$ but its closed form is not determined here; identifying it (and whether it relates to $\ln(4/3)$, $\ln 3$, or the Euler–Mascheroni constant) is left open. No claim of a new transcendental constant is made; the earlier value 0.9398 is accounted for as a finite-q effect.

12 Extension of Tao's Almost All Result

Tao's 2019 theorem [5] establishes: for any $f : N \to R$ with $\lim_{n\to\infty} f(n) = +\infty$, the set $\{n : \mathrm{Col}_{min}(n) > f(n)\}$ has logarithmic density zero. More precisely,

$$\lim_{x\to\infty} \frac{1}{\log x} \sum_{\substack{n<x \\ n\in E}} \frac{1}{n} = 0, \qquad (37.)$$

where E is the exceptional set.

Remark 12.1 (Scope of hypergeometric extension). *The Riemann P-equation framework does not, by itself, extend Tao's theorem to a stronger density result. What it provides is:*

1. *A continuous geometric interpretation of Collatz orbits, via the lifting of the Möbius maps $\mathcal{M}_E(\varphi)$,$\mathcal{M}_O(\varphi)$ to monodromy matrices of a Fuchsian ODE. This gives an analytic framework within which Tao's probabilistic estimates can be reinterpreted as statements about the monodromy spectrum.*

2. *A non-resonance mechanism: parametrizing exponents as $\alpha = n/2 + \phi$ with ϕ irrational ensures that exponent differences are never integers, so monodromy matrices are diagonalizable (no Jordan blocks) and orbits are quasiperiodic rather than periodic. This provides a heuristic explanation for why rational resonances (which could cause periodic cycles) do not appear for generic starting integers.*

3. *The constant β equals 1 exactly, so the digamma-averaged framework reproduces the classical growth constant with no systematic rescaling; the apparent finite-size value near 0.94 is explained as a slow 1/ln q correction. This serves as a consistency check on the framework rather than a new constant.*

No claim is made to have proved that the exceptional set has Hausdorff dimension zero or that the density extends to full Lebesgue measure; these remain open.

13 Kummer's 24 Solutions and Analytic Continuation

The 24 solutions to the Riemann P-equation (Kummer's 24 solutions, see [2]) arise from 6 permutations of the singularities $(0,1,\infty)$ and 4 choices of exponent pairs at each singularity. The Euler and Pfa transformations, e.g.,

$$2\ {}_2F_1(a,b;c;z) = (1 - z)c - a - b\ {}_2F_1(c - a, c - b; c; z), \tag{38.}$$

generate this set.

When exponent differences are non-integers (as when $\alpha = n/2 + \phi$ with ϕ irrational), solutions are linearly independent power series without logarithmic terms. The ratio of two solutions sharing a common power-law prefactor,

$$r(z) = \frac{f_1(z)}{f_2(z)} = \frac{{}_2F_1(\alpha,\beta;\gamma;z)}{z^{1-\gamma}\ {}_2F_1(\alpha-\gamma+1,\beta-\gamma+1;2-\gamma;;z)} \tag{39.}$$

has effective leading exponent $\gamma - 1$, which is non-integer by assumption. This ratio satisfies a derived (typically first-order) ODE, simplifying the dynamics and filtering resonances. In the Collatz context, this corresponds to subtracting the immediate-collapse terms (integer parts of the exponent), leaving non-integer residuals that capture delayed collapse without amplifying fluctuations.

14 Collapse Theorem via Digamma Cotangent Contraction

De nition 14.1. For $\Phi \in (0,1)$, de ne the digamma cotangent potential

$$U(\Phi) := \psi(1 - \Phi) - \psi(\Phi) = \pi\cot(\pi\Phi) \tag{40.}$$

The unique zero of U on $(0,1)$ is $\Phi = \frac{1}{2}$, corresponding to $n = 1$.

Theorem 14.1 (Layer-tuned collapse to $\Phi = \frac{1}{2}$). *Fix $n_0 \in N$ and set $\Phi_0 = \frac{1}{n_0+1}$. Define the contraction factors*

$$\rho_0(n) = \frac{1}{2n+2}, \qquad \rho_E(n) = \frac{2}{n+2} \tag{41.}$$

and the update rule on

$$U(\Phi_{k+1}) = \begin{cases} \rho_0(n_k)\Phi_k & n_k \ \ odd, \\ \rho_E(n_k)\Phi_k & n_k \ \ even, \end{cases} \tag{42.}$$

Then $\rho_0(n_k) \le \frac{1}{4}$*, and* $\rho_E(n_k) \le \frac{1}{2}$ *for all* $n \ \ge \ 1$*, so* $U(\Phi_{k+1}) \le \frac{1}{2}|\Phi_k|$ *at every step, giving*

$$|U(\Phi_k)| \le \left(\frac{1}{2}\right)^k |\, U\Phi_0| \to \ 0 \tag{43.}$$

and since U *is continuous on* $(0,1)$ *with unique zero at* $\Phi = \frac{1}{2}$*,*

$$\Phi_k \to \frac{1}{2} \quad as\ k \to \infty \tag{44.}$$

Proof. For $n \ \ge \ 1$:

$$\rho_0(n) = \frac{1}{2n+2} \le \frac{1}{2\cdot 1+2} = \frac{1}{4} < \frac{1}{2} \tag{45.}$$

For $n \ \ge \ 2$ even (the even step applies only when n is even, so $n \ \ge \ 2$):

$$\rho_E(n) = \frac{1}{n+2} \le \frac{1}{2+2} = \frac{1}{2} \tag{46.}$$

Hence at every step, regardless of parity, $|U(\Phi_{k+1})| \le \frac{1}{2}|U(\Phi_k)|$. By induction, $|U(\Phi_k)| \le 2^{-k}|U(\Phi_0)|$. As $k \ \to \ \infty$, $U(\Phi_k) \to \ 0$. Since $U(\Phi) \ = \ \pi \cot(\pi\Phi)$ is continuous and strictly monotone on $(0,1)$ with $U\left(\frac{1}{2}\right)$=0, the convergence $U(\Phi_k) \to 0$ implies $\Phi_k \to \frac{1}{2}$.

Remark 14.1 (Relationship to the Collatz conjecture). Theorem 14.1 proves convergence of the abstract contraction system defined by ρ_0, ρ_E. It does not prove the Collatz conjecture because the connection between the abstract contraction factors and the actual Collatz steps requires establishing that every orbit stays in the domain where ρ_0, ρ_E apply. Specifically, the theorem assumes no orbit reaches $n = 0$ or escapes to $n \ = \ \infty$; proving this for all starting points is precisely the content of the Collatz conjecture. Theorem 14.1 should be understood as: if the orbit stays bounded, then it converges to $n \ = \ 1$, which is a conditional statement consistent with, but not a proof of, the conjecture.

14.1 Block map and the log-energy drift condition

Writing $3n \ + \ 1 \ = \ 2^{m_j}n'$ with n' odd and $m_j \ = \ \nu_2(3n \ + \ 1)$, the closed-form block map (one odd step followed by m even steps) in Φ-coordinates is

$$E^m(\xi) = \frac{2^m}{1+(2^m-1)\xi} \tag{47.}$$

After one odd step from Φ, then m even steps:

$$\Phi\overrightarrow{\text{odd} + m \text{ even}} \quad \Phi^{(\mathrm{m})} = \frac{2^m}{3 + (2^m - 2)\Phi} \tag{48.}$$

The log-energy $L = \log U(\Phi) \approx logcot(\pi\Phi)$ drifts by

$$\Delta L = \log 3 - m_j \log 2 \tag{49.}$$

The sufficient condition for collapse is:

$$\sum_{j=1}^{\infty} m_j \log 2 - \log 3 = +\infty \text{, or equivalently,} \quad \liminf_{\mathcal{T}\to\infty} \frac{1}{\mathcal{T}} \sum_{j=1}^{\mathcal{T}} m_j > \log_2 3 \approx 1.585 \tag{50.}$$

Under the standard heuristic the average $\frac{1}{\mathcal{T}}\sum m_j$ converges to 2 (the expected 2-adic valuation of $3n + 1$ under the heuristic distribution of odd n), which for almost all starting integers (in logarithmic density) gives a mean valuation exceeding $\log_2 3 \approx 1.585$, thereby satisfying the sufficient collapse condition stated above.

15 Conclusions

This paper developed a two- field geometric model on the integer plane in which odd and even layers carry, respectively, an expansive vector eld (slope $2n + 1$) and a collapsive eld (slope $-n/2$). The principal contributions of this paper are:

1. **Switching rules**. The even-field ODE$\dot{n} = -\frac{n}{2}$ integrates to$n(x) = n_1 e^{-x/2}$, implementing halving at flight time $T_e = 2\nu_2(n_1)\ln 2$. The odd- eld ODE $\dot{n} = 2n + 1$ integrates to $n(x) = \left(n_0 + \frac{1}{2}\right)e^{2x} - \frac{1}{2}$, implementing $n \to 3n + 1$ at the universal flight time $T_0 = \frac{\ln 3}{2}$.

2. **Φ-coordinate Möbius formulation.** Under the embedding $\Phi(\mathrm{n}) = \frac{1}{\mathrm{n}+1}$ (carefully distinguished from the digamma function ψ), the Collatz map becomes two Möbius transformations $\mathcal{M}_E(\varphi)$,$\mathcal{M}_O(\varphi)$, making the connection to the Riemann P-equation's Möbius symmetries (5) precise.

3. Formal equivalence proposition. Proposition 6.1 establishes the precise and limited sense in which the Collatz Möbius action is analogous to (not realized by) the monodromy of a Riemann P-equation: both act on a projective coordinate by Möbius transformations, the monodromy matrices acting on the ratio of solutions by the same form as $\mathcal{M}_E(\varphi)$,$\mathcal{M}_O(\varphi)$.

4. Strengthened theorems. Theorem 8.1 (no in infinite expansion) is proved rigorously by 2-adic induction. Theorems 10.3 and 10.4 establish the harmonic density structure and the Euler Mascheroni limit, with complete proofs.

Collatz Digamma constant. The shifted digamma average satisfies $\beta = lim\,(b(q) + ln\, q) = 1$ exactly as $q \to \infty$, with the leading term established via the elementary identity $\int_0^1 ln(1-x)\,dx = -1$. The convergence is governed by a slow $O(1/ln\, q)$ correction, so a finite-q computation reads off an apparent value near 1 (e.g., near q ~ 2×10^5); this is a finite-size way-station, not the limit. Since β = 1, the effective growth rate equals the classical heuristic constant $c = 2/\ln(4/3) \approx 6.952$ with no systematic reduction. The framework reproduces the classical constant rather than modifying it (a consistency check).

5. Collapse theorem (conditional). Theorem 14.1 proves that the abstract digamma cotangent system contracts geometrically to $\Phi(\mathrm{n}) = \frac{1}{2}$. The connection to the full Collatz conjecture is conditional: the theorem shows convergence given that orbits remain bounded.

What remains open. The Collatz conjecture itself is not proved. The Riemann P-equation link is an analogy of form, not of kind (Remarks 5.2 and 6.1): the Collatz Möbius generators are hyperbolic with multipliers 2 and 3, whereas hypergeometric monodromy is elliptic, so there is no literal realization. The extension of Tao's logarithmic density result to fuller measures is not established; it remains a heuristic motivation. These are directions for future work.

16 Relation to prior work

The results of this paper sit within a well-developed body of work on the Collatz (Syracuse) map, and it is worth stating precisely what is used, what is recovered, and what is not extended. The stopping-time and density framework employed originates with the late-1970s analytic work on the generalized Syracuse problem, including Allouche [8], and was organized into its modern form by the survey of Lagarias [6], whose vocabulary of stopping times, total stopping times, and the dynamical framing of the map underlies our Section 10. The logarithmic growth of the mean stopping time established there, $A(q) \sim K \ln q$ with $K = 3/\ln(4/3)$, is governed by the same per-step log-drift $\ln(4/3)$ that fixes Korec's bounds on $3n + 1$ sequence lengths [7]; the classical average-stopping-time constant ≈ 9.478 that our numerics approach is a known quantity in this literature, not a new one. The analysis recovers it as a consistency check rather than claiming an improvement.
Our harmonic and digamma results are averaged, heuristic-level statements about the aggregate structure of orbits; they are of a different character from the rigorous density theorems of this tradition, and does not sharpen those. In particular, the strongest unconditional results on how far orbits descend, Korec's natural-density estimate that almost all n reach a value below n^{θ} for any $\theta > \log_4 3 \approx 0.7924$, and Tao's logarithmic-density theorem [5] that almost all orbits eventually fall below any prescribed $f(N) \to \infty$—are not extended by the present framework. Our Section 12 uses Tao's result only as motivation and consistency: the two-field model reproduces the log-drift picture underlying it, but does not upgrade logarithmic density to natural density, nor "below $f(N)$" to convergence to 1. Theorem 14.1 is explicitly conditional at exactly this point.
Where the present work adds structure rather than density estimates is in the projective reformulation. The Φ-coordinate realization of the Collatz map as a semigroup of two Möbius generators (Proposition 5.1), and their classification as hyperbolic isometries with multipliers equal to the two arithmetic primes 2 and 3 (Remark 5.2), give a concrete $\mathrm{PGL}(2, \mathbb{Q})$ representation of the multiplicative structure studied in the semigroup approach to the problem. The 2-adic non-expansion result (Theorem 8.1)—that long $m = 1$ runs occur exactly at $n_0 = 2^k - 1$, converging to the 2-adic integer -1—reflects the eventually-periodic binary structure familiar from the automatic-sequence and 2-adic viewpoint on such maps. These are structural observations that complement, rather than compete with, the density-theoretic results above.
In summary: Lagarias [6] and Allouche [8] supply the framework; Korec [7] and Tao [5] represent the rigorous descent results, which are consistent with the paper's claims, with but do not extend; and the contributions of this paper-the Möbius/monodromy structure, the hyperbolic-multiplier classification, the 2-adic non-expansion theorem, and the $\beta = 1$ consistency check, are structural rather than density-improving.

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